

AI Data Quality Analysis Report

Comprehensive Synthetic Data Validation

Dataset Information:

Real Dataset: 10000 rows, 10 columns

Synthetic Dataset: 10000 rows, 10 columns

Analysis Date: 06/01/2026

Executive Summary

The comprehensive validation of synthetic diabetes dataset reveals a complex scenario of data generation with nuanced implications. The synthetic data demonstrates an exceptional level of statistical alignment with the original dataset, showing perfect replication across multiple validation dimensions. With 50 total tests conducted, 25 passed tests, and zero critical failures, the synthetic data presents a remarkable mirror of the original dataset.

Key statistical observations include:

- 100% column compatibility between real and synthetic datasets
- Identical statistical distributions across all variables
- Perfect correlation matrix preservation
- No significant differences detected in range, distribution, or statistical tests

However, the exact match rate of 1.0 (10,000/10,000 rows) raises potential concerns about over-fitting or potential privacy risks. While this suggests precise data reproduction, it might compromise the fundamental goals of synthetic data generation, which typically aim to preserve statistical properties while introducing controlled variability.

The dataset originates from a diabetes-related medical context, containing 10 variables including patient characteristics like pregnancies, glucose levels, blood pressure, BMI, age, and diabetes outcome. The validation process employed rigorous statistical techniques including Kolmogorov-Smirnov tests, correlation comparisons, Welch's t-tests, and multivariate distribution assessments to ensure comprehensive data quality evaluation.

The overall data quality is characterized as "POOR" despite perfect statistical alignment, suggesting potential underlying methodological limitations in the synthetic data generation process.

Key Findings

The validation results present a fascinating paradox of data replication with significant methodological implications:

Statistical Alignment:

- 100% column compatibility between real and synthetic datasets
- Identical mean and standard deviation across all 10 variables
- Zero statistically significant differences in distribution tests
- Perfect preservation of correlation matrix (45 total correlation comparisons)

Outlier Characteristics:

- Several variables exhibit notable outlier patterns:

1. BloodPressure: 5.37% outliers
2. Insulin: 4.4% outliers
3. DiabetesPedigreeFunction: 3.74% outliers
4. BMI: 2.6% outliers
5. Age: 1.69% outliers

Distribution Tests:

- Kolmogorov-Smirnov tests consistently returned p-values of 1.0
- Welch's t-tests showed zero statistical differences
- Multivariate tests like total variation and KL divergence tests returned "ACCEPT" results

Privacy and Utility Metrics:

- Data completeness: 100% for both real and synthetic datasets
- Nearest Neighbor Distance Ratio: Median and mean of 0
- Exact Match Rate: 100% (10,000/10,000 rows)

Critical Observation:

The 100% exact match rate is unprecedented and potentially problematic. While statistically perfect, this suggests the synthetic data generation method may have simply copied the original dataset rather than creating a genuinely synthetic representation with preserved statistical properties.

Statistical Quality

The statistical quality assessment reveals a meticulously replicated dataset with extraordinary precision:

Distribution Consistency:

- All 10 variables demonstrate identical statistical distributions
- Kolmogorov-Smirnov tests consistently returned p-value of 1.0
- Zero statistically significant differences detected

Correlation Structure:

- 45 total correlation comparisons
- 100% correlation matrix preservation
- Jennrich test for correlation matrix comparison returned "ACCEPT" result

Variance and Moments:

- Mean values perfectly preserved across all variables
- Standard deviations identical for each variable
- Skewness and kurtosis characteristics maintained

Multivariate Tests:

- Total Variation Test: Statistic of 0, non-significant
- KL Divergence Test: Statistic of 0, non-significant
- Jennrich Test: Successful correlation matrix comparison

Outlier Characteristics:

- Consistent outlier patterns across real and synthetic datasets
- Outlier percentages range from 0% to 5.37%
- No statistically significant differences in outlier distributions

Limitations:

The 100% statistical alignment suggests potential methodological limitations in the synthetic data generation process. While statistically perfect, this may indicate:

1. Potential overfitting
2. Lack of genuine synthetic data generation
3. Risk of inadvertent privacy breach

The validation suggests an urgent need to review and potentially modify the synthetic data generation methodology to introduce controlled variability while maintaining statistical fidelity.

Practical Usefulness

Practical usefulness assessment reveals a nuanced perspective on the synthetic dataset's applicability:

Data Utility Dimensions:

- 100% data completeness
- Identical statistical distributions
- Preserved correlation structures
- Consistent variable ranges

Potential Use Cases:

1. Preliminary model training and testing
2. Educational and demonstration purposes

3. Initial algorithm development

Limitations:

- Exact 100% match raises significant reproducibility concerns
- Potential privacy risks due to near-perfect replication
- Limited generalizability due to lack of introduced variability

Contextual Considerations:

- Medical/diabetes research domain requires high fidelity
- Synthetic data should preserve statistical properties
- Current implementation lacks controlled stochastic variation

Privacy Preservation Assessment:

- Nearest Neighbor Distance Ratio: 0
- Exact Match Rate: 100%
- Potential re-identification risks

Recommendation Framework:

- Introduce controlled noise
- Implement differential privacy techniques
- Develop more sophisticated synthetic data generation algorithms

Practical Scoring:

- Data Utility: Moderate
- Privacy Preservation: Low
- Generalizability: Limited

The synthetic dataset demonstrates technical precision but fails to achieve the fundamental objectives of synthetic data generation: statistical preservation with controlled variability and enhanced privacy.

Critical Issues

Critical issues identified in the synthetic data validation:

1. Exact Replication Concern

- 100% exact match rate (10,000/10,000 rows)
- Suggests potential direct copying rather than genuine synthetic generation
- Severe privacy and reproducibility implications

2. Lack of Controlled Variability

- Identical statistical moments across all variables
- No introduced stochastic variation
- Compromises core synthetic data generation principles

3. Potential Privacy Risks

- Perfect correlation matrix preservation
- Zero variation in nearest neighbor distances
- High risk of re-identification

4. Methodological Limitations

- Energy test returned "ERROR"
- Suggests underlying computational or algorithmic constraints
- Indicates potential implementation challenges

5. Outlier Preservation

- Consistent outlier patterns across variables
- Potential information leakage
- Reduces anonymization effectiveness

Immediate Recommended Actions:

- Redesign synthetic data generation algorithm
- Implement differential privacy techniques
- Introduce controlled statistical noise
- Enhance privacy preservation mechanisms

The identified critical issues underscore the need for a comprehensive reassessment of the synthetic data generation methodology.

Variable Range and Type Checks

Variable Range and Type Analysis:

Numeric Variables with Potential Concerns:

1. Glucose

- Range: -78.63 to 224.32
- Negative values unlikely in physiological context

- Potential data preprocessing or generation artifact

2. BloodPressure

- Range: -31.30 to 121.24
- Negative values physiologically impossible
- Indicates potential data generation or transformation issue

3. Insulin

- Range: -12.16 to 889.23
- Negative values not medically meaningful
- Extremely high maximum value requires investigation

4. BMI

- Range: -20.57 to 61.12
- Negative values invalid for Body Mass Index
- Suggests potential normalization or scaling problem

5. DiabetesPedigreeFunction

- Range: -0.017 to 3.20
- Negative values unexpected
- Requires domain-specific validation

Data Type Consistency:

- All variables maintain consistent numeric type
- No type mismatches detected between real and synthetic data

Recommendations:

- Review data preprocessing steps
- Implement range validation during data generation
- Ensure physiologically meaningful value ranges
- Consider domain-specific constraints in synthetic data generation

Real Data Quality by Variable

Real Data Variable Quality Assessment:

1. Pregnancies

- Range: -1.36 to 16.55
- Negative values impossible
- Mean: 3.97
- Potential data collection or preprocessing issue

2. Glucose

- Range: -78.63 to 224.32
- Physiologically improbable negative values
- Mean: 122.65
- Requires careful domain validation

3. BloodPressure

- Range: -31.30 to 121.24
- Negative values medically nonsensical
- Mean: 69.11
- Suggests potential measurement or transformation errors

4. Insulin

- Range: -12.16 to 889.23
- Extreme variability
- Mean: 83.24
- High standard deviation (118.49)
- Potential outliers or measurement inconsistencies

5. BMI

- Range: -20.57 to 61.12
- Negative values invalid
- Mean: 32.04
- Requires careful preprocessing

6. DiabetesPedigreeFunction

- Range: -0.017 to 3.20
- Small range with negative values
- Mean: 0.477
- Requires domain-specific interpretation

Recommendations:

- Conduct thorough data cleaning
- Implement robust outlier detection
- Validate data collection methodology
- Consider domain expert consultation for range validation

Actionable Recommendations

Comprehensive Recommendations for Synthetic Data Generation:

1. Enhance Privacy Protection

- Implement differential privacy techniques
- Introduce controlled statistical noise
- Develop probabilistic data generation methods
- Use k-anonymity and l-diversity principles
- Randomize synthetic data generation seed

Justification: Current 100% match rate poses significant re-identification risks.

2. Improve Statistical Variability

- Develop stochastic generation algorithms
- Introduce controlled randomness
- Preserve statistical moments while allowing variation
- Implement advanced generative models like GANs
- Use domain-specific constraints

Justification: Existing method lacks genuine synthetic data characteristics.

3. Validate Physiological Constraints

- Implement domain-specific range validation
- Remove physiologically impossible values
- Enforce medically meaningful ranges
- Collaborate with domain experts
- Develop robust preprocessing pipelines

Justification: Current data contains negative values in health-related variables.

4. Develop Comprehensive Validation Framework

- Create multi-dimensional validation metrics
- Implement advanced statistical testing
- Develop privacy risk assessment protocols
- Design iterative validation processes
- Establish synthetic data quality benchmarks

Justification: Current validation process reveals methodological limitations.

5. Advanced Generative Modeling

- Explore machine learning generative techniques
- Investigate variational autoencoders
- Develop hybrid generative approaches
- Integrate domain knowledge into generation process
- Continuously refine generation algorithms

Justification: Existing method demonstrates technical precision but lacks fundamental synthetic data generation principles.

Risk Level

Risk Level: MEDIUM

Justification:

- 100% exact data match indicates high privacy risks
- Lack of controlled variability compromises data utility
- Physiologically impossible values in critical health variables
- Potential re-identification vulnerabilities
- Methodological limitations in synthetic data generation

The medium risk classification stems from the technical precision combined with significant structural and privacy-related concerns that require immediate methodological refinement.

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